1 Introduction

1.1 Aim

To study γ absorption in matter by γ-ray source

- To find attenuation constant for aluminium.
- To verify inverse sqaure law of radiation.

1.2 Brief theory

Geiger-Muller device has has two parts, namely GM tube and Electronic Counter.

1.2.1 GM Tube

The Geiger Muller Tube is a gaseous ionization detector and it uses the Townsend Avalanche phenomenon to detect ionizing radiation. It has cathode and an anode with a couple hundred volts set as the potential difference. The medium in between is filled with atmospheric gas at a very low pressure of 0.1 atm. When ionizing radition comes into this medium, it strikes off an electron of the gas molecules, thus creating a ion-electron pair.

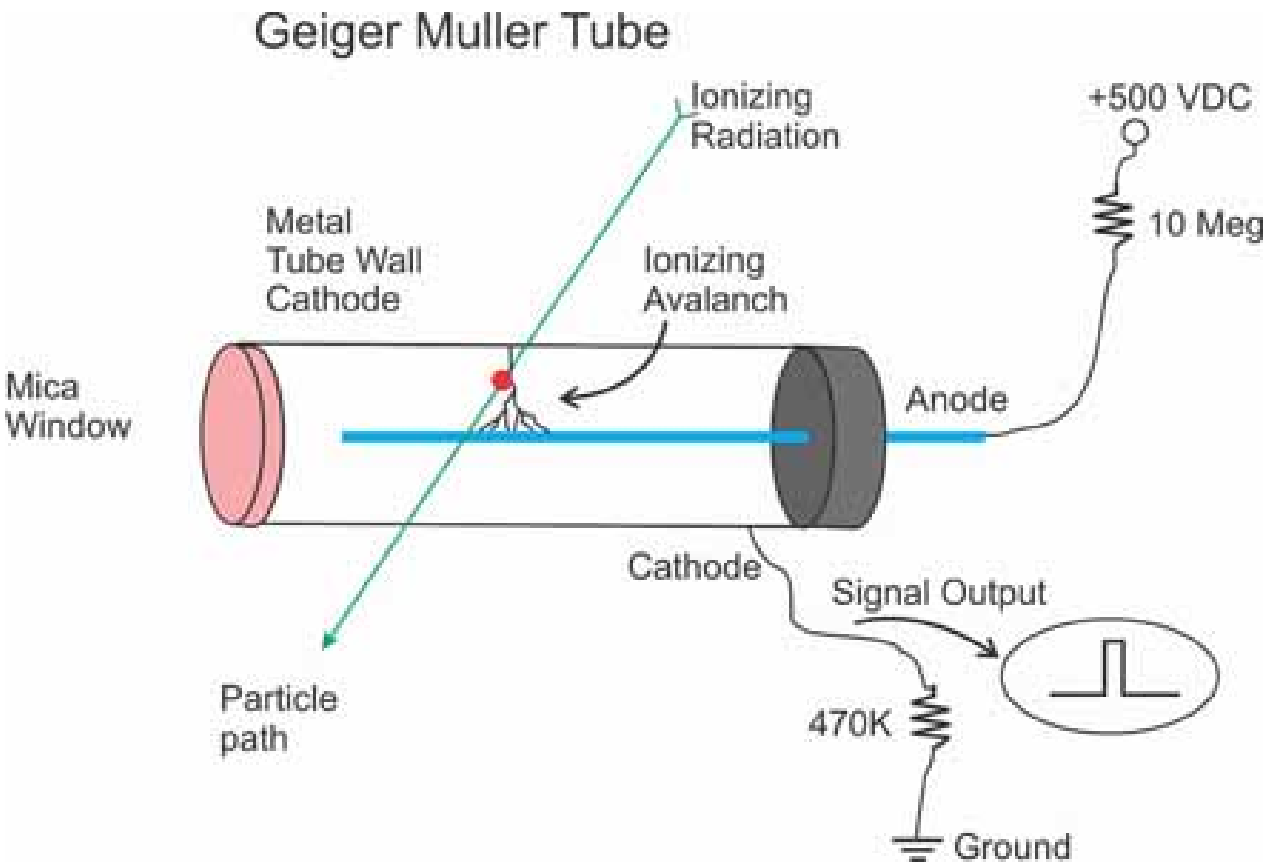


Figure 1: Scematics of a typical Gieger-Muller tube

Due to the high potential difference between the cathode and the anode the electron gets accelerated and gains suffiecient energy to knock off further electrons of other gas molecules and a kind of chain reaction or avalanche take place. The avalanche gives rise to the gas multiplication effect which is

responsible for production of a significant output from a single ionizing event. If there is a single event then a single avalanche results in producing $10^6 - 10^8$ ion-electron pairs. However the production of UV photons at the original avalanche starting point which is uncharged and unaffected by the electric field and can travel laterally to the Electric Field) generates new electronion pairs thereby making multiple avalanches.

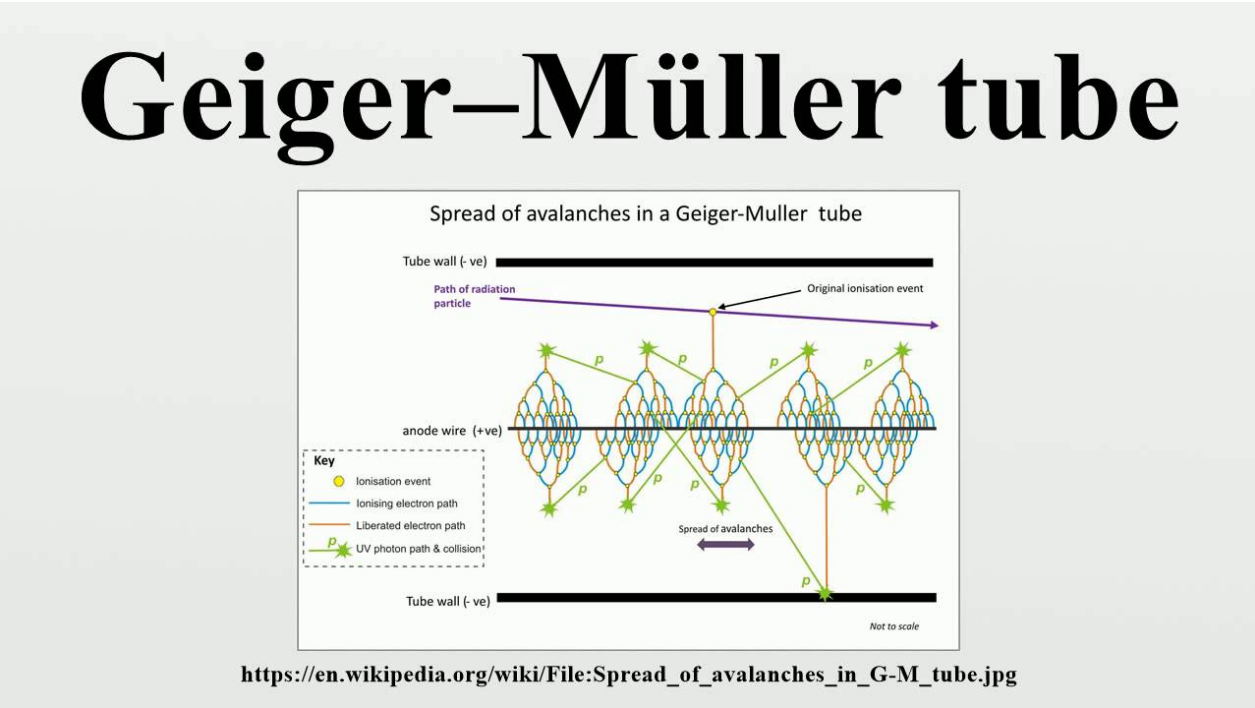


Figure 2: spread of avalanche in gm tube

Multiple avalanches increase the number of such pairs to $10^9 - 10^{10}$. The avalanches proceed at a speed of 2 to 4 $cm/\mu sec$. So a single event produces about 10^9 elctrons which is received by the anode in a very short duration of time 2-4 $\mu secs$. This creates a current pulse which can be detected as a voltage pulse against a resistor. Such a voltage pulse is termed a count which stands for a single ionizing event. Such a voltage pulse is termed a count which stands for a single ionizing event. Note that the current produced is a function of the number of avalanches produced which in turn depends on the geometry of the detector, the electric field strength and type of gas. Hence no information about radiation energy can be found. Once a avalanche is produced it creates a lot ions which are relatively heavier and accumulate near the anode. This counters the electric field which stops the avalanche generation and the discharge is terminated. Sometimes when the ions get collected on the cathode it is in a higher state of energy and hence releases a high energy photon which can trigger secondary avalanches. This gives spurious pulses which needs to be controlled. This is done by using special types of gases (chemical quenching) and high speed electronics to remove the electric field after a pulse generation (external quenching). Note that the GM counter should be operated in the Geiger plateau region voltage. The Geiger plateau is the voltage range in which the GM tube operates in its correct mode. If a GM tube is exposed to a steady radiation source and the applied voltage is increased from zero, it follows the plot of ion current shown in this article. In the Geiger region the gradient flattens; this is the Geiger plateau.

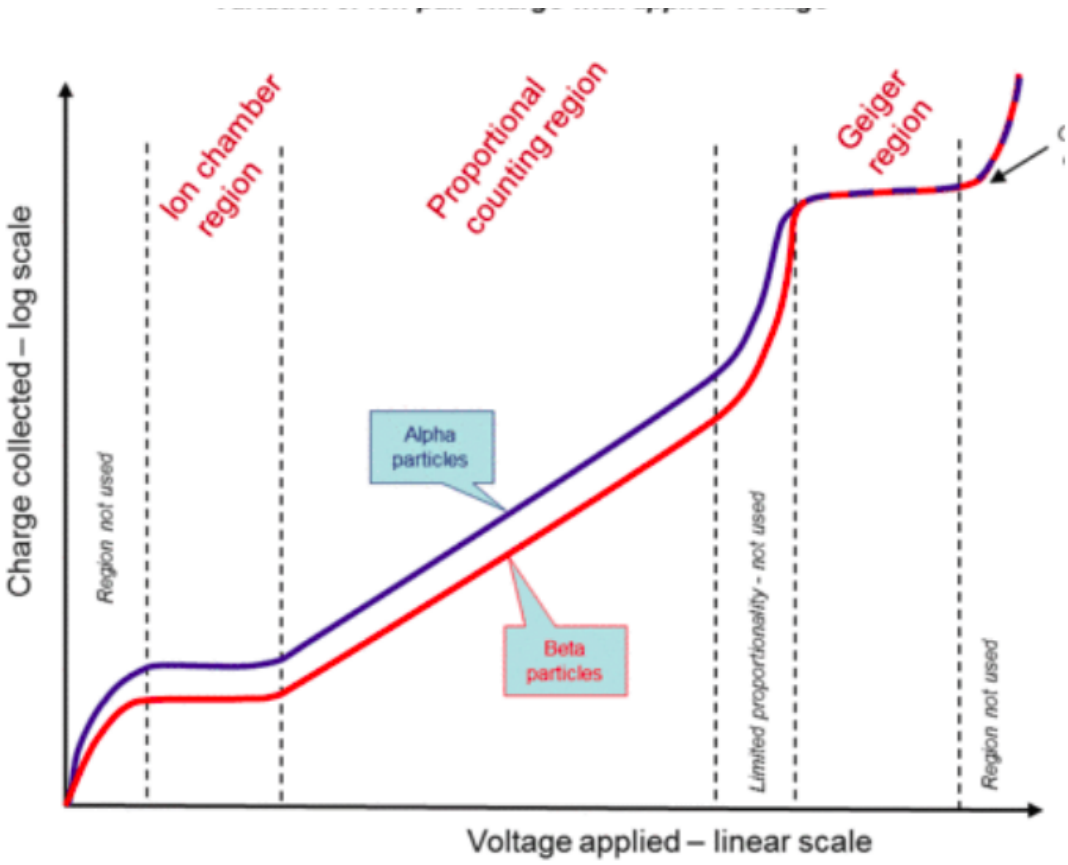


Figure 3: spread of avalanche in gm tube

1.2.2 Electronic counter

Via this mechanism, the apparatus is able to store the counts of pulses. Of each incoming pulses, the counter converts it into square wave, stores it in memory. The other important function of it is the Self-suppressing Mechanism. After a pulse is registered, the counter feeds back a opposite polarity square pulse of high enough amplitude and of duration 300-500 μsec to the central wire, in order to stop the discharge. Thus, the tube resets as fast as possible to register next pulse.

1.2.3 Dead time of GM tube

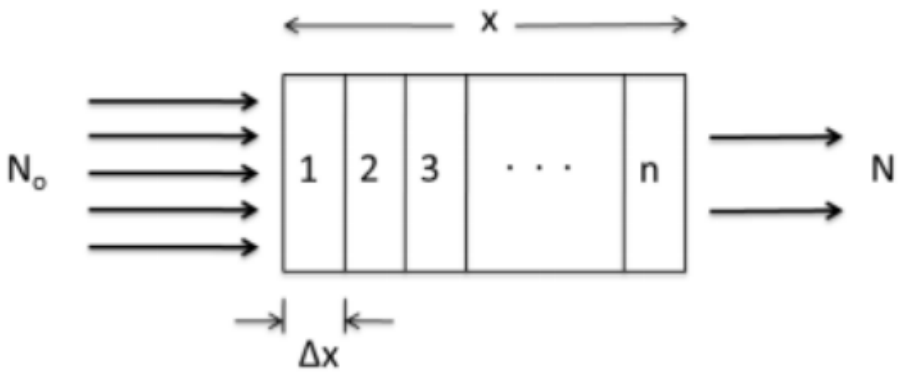
Due to time required for recombination of ions after an ionizing event, GM tube exhibits Dead Time. The actual dead time depends on lots of factors (e.g. active volume and shape of the detector, rate of incoming radiation, nature of the gas filled etc.). It can vary from few $\mu secs$ for small tubes to few milisecs for larger tubes. As the discharge occurs very close to the wire and due to massiveness of the ions relative to electrons, while electrons move fast to the central wire (assuming it is the anode), the ions move much slower. Hence, the +vely charged ions form a sheath around positively charged wire. This reduces the effective electric field in the tube and if the value is lower than the threshold then the counter cannot register anymore pulses. The ions gradually recede towards cathode, when after a critical radius, r, the field is again restored. The time required for this whole event is Dead Time.

1.2.4 Absorption of gamma rays

The gamma ray can be absorbed by a given substance by following processes

- The photoelectric effect
- Compton scattering
- pair production

In all three of these processes the gamma ray is either scattered away from the incident direction or completely absorbed. That is, a detector is placed on the opposite side of the absorber, along the incident direction of a beam of gamma rays, only those gamma rays which did not interact with the absorber will be detected. Wheather the gamma ray will interact with matter by any of above mentioned proccess is probabiltist thus the gamma ray absorption is a random type of process. It is not possible to say whether a particular gamma ray will interact with the absorber or pass through unaffected. Let N_0 be number of number of gamma rays incident on an absorber of thickness x. We can devide the absorber into n sections of equal thickness Δx.



We can assume that probability that a gamma ray is abosrbed is praportional to number of particle incident and the thicknes. For small thickness Δx we can assume that number of incident particle N is same for entire Δx. Thus

$$\Delta N \propto -N\Delta x \tag{1}$$

We define praportionality constant to be linear attenuation constant (μ) then eqation(1) becomes

$$\Delta N = -\mu N\Delta x \tag{2}$$

In the limit N $\rightarrow \infty$ Δ can be replaced with differential and from equation(2) we get

$$\frac{dN}{N} = -\mu dx$$

$$\implies \int_{N_0}^{N} \frac{dN}{N} = \int_0^x -\mu dx$$

$$\implies N = N_0 e^{-\mu x}$$

Gamma rays interact primarily with atomic electrons. Hence the attenuation coefficient must be proportional to the electron density P, which is proportional to the bulk density of the absorbing material. For a given material the ratio of the electron density to the bulk density is a constant and is equal to Z/A. Where Z is atomic number and A is atomic mass. Thus P = $\frac{Z\rho}{A}$. The ratio of the linear attenuation coefficient to the density ($\frac{\mu}{\rho}$) is called the mass attenuation coefficient.

1.2.5 Inverse square law of radiation

In absence of aluminium plate the absorption is negligible, gamma rays follow an inverse square law because the same power is spread out in an area of $4\pi r^2$ so the intensity goes down as $\frac{1}{r^2}$.Thus if we plot he log(I) vs (logr) then we should get a straight line with slope of -2. However due to scattering, absorption from air molecules it will not exactly be -2.

2 Data

2.1 Raw data

Gamma ray source: Na^{22}
We find counts in the absence of Na^{22}, which is the background reading. The data was taken for 10 minutes and counts obtained are 222 and 271. Average of these two data is 0.42 count per second.
We measure the thickness of aluminium foils using screw gauge. For this zero error is 0. Least count on linear scale is 0.5cmm. Least count of circular scale is 0.01mm. Now we measure the count for interval of 30 seconds at different thickness of aluminium foils. This data is presented in table-1. Correction is required as we have to subtract the background.

Thickness (mm)	**counts**	**Average**	**Corrcted counts**	**Thickness (mm)**	**counts**	**Average**	**Corrcted counts**	**Thickness (mm)**	**counts**	**Average**	**Corrcted counts**
0.48	83	89	77	13.24	58	69	56		67		
	80				69				68		
	106				80			26.71	56	63	51
0.98	85	88	75	13.72	74	72	59		76		
	83				63				59		
	97				79			27.2	68	62	49
1.47	94	95	83	14.2	81	71	59		61		
	97				59				58		
	96				75			27.68	63	56	43
1.96	101	82	70	14.92	72	71	58		56		
	70				82				50		
	77				60			28.07	66	69	57
2.44	71	75	62	15.4	67	69	56		82		
	85				65				61		
	70				75			28.55	56	64	51
2.93	92	96	84	15.89	61	69	56		68		
	111				68				68		
	87				79			29.04	62	70	57
3.42	80	88	76	16.37	78	75	62		69		

	99				71				79		
	87				76			29.52	67	69	57
3.92	84	88	75	16.86	77	69	56		78		
	97				57				64		
	84				73			30.01	57	59	46
4.4	83	78	65	17.34	71	67	54		56		
	73				59				64		
	79				72			30.49	70	60	48
4.9	76	84	71	17.83	61	71	58		52		
	85				84				60		
	91				69			30.98	65	62	50
5.57	68	74	61	18.31	54	63	51		52		
	78				60				71		
	76				77			31.46	57	66	53
6.15	102	83	70	18.72	68	64	51		64		
	81				65				77		
	66				59			31.95	53	52	39
6.79	85	82	69	19.2	79	69	56		45		
	87				60				58		
	75				69			32.43	59	56	44
7.27	71	81	69	19.69	54	66	53		51		
	90				70				60		
	84				74			32.92	51	62	49
7.76	86	80	68	20.17	59	62	50		78		
	82				69				58		
	74				60			33.64	64	52	40
8.23	79	80	68	20.66	85	75	63		54		
	88				70				40		
	75				72			33.12	59	62	49
8.73	67	76	63	21.14	60	62	49		62		
	87				61				65		
	74				66			34.61	55	56	43
9.21	70	81	68	21.63	54	56	44		60		
	82				64				53		
	92				52			35.09	68	60	47
9.84	80	78	65	22.11	58	55	42		55		
	74				53				57		
	81				54			35.58	59	55	43
10.33	77	74	62	22.6	54	66	53		48		
	76				66				60		
	71				78			36.06	52	58	45
10.81	73	83	70	23.08	57	57	44		60		
	85				60				62		
	92				54			36.55	57	64	52
11.3	75	76	63	23.56	63	66	53		62		
	90				57				75		

	64				78			37.03	56	58	46
11.78	77	75	63	24.29	57	63	51		64		
	86				64				56		
	64				70			37.52	56	55	43
12.27	71	86	73	25.26	69	69	56		52		
	105				53				59		
	82				85			38	67	62	49
12.75	91	79	67	25.74	67	66	54		60		
	72				58				59		
	76				75						
				26.23	54	54	41				

Table 1: Number of counts for 30 sec from a gamma source for different thickness of Aluminium sheet

Now we remove the aluminium sheet and measure the count by varying the distance between the source and the detector. The count data for different distance is presented in table-2.

Distance (cm)	**Actual Distance (cm)**	**Time (sec)**	**Counts**	**Counts/ second**	**Average Counts/ second**	**Corrected Count**
7.5	7.98	60	1161	19	18	18
			1192	19		
			1107	18		
8	8.48	60	992	16	16	16
			1045	17		
			1004	16		
8.5	8.98	30	428	14	15	14
			482	16		
			470	15		
9	9.48		388	12	12	12
			406	13		
			397	13		
9.5	9.98		366	12	11	11
			333	11		
			344	11		
10	10.48		293	9	9	9
			320	10		
			306	10		
10.5	10.98		293	9	9	9
			286	9		
			303	10		
11	11.48		271	9	8	8
			264	8		
			226	7		
11.5	11.98		245	8	7	7
			236	7		

			223	7		
12	12.48		224	7	6	6
			198	6		
			211	7		
12.5	12.98		187	6	6	6
			196	6		
			199	6		
13	13.48		180	6	5	5
			167	5		
			158	5		
7	7.48		618	20	20	20
			604	20		
			623	20		
6.5	6.98		744	24	25	25
			798	26		
			765	25		
6	6.48		940	31	29	28
			841	28		
			850	28		
5.5	5.98		993	33	34	33
			1033	34		
			1051	35		
5	5.48		1281	42	42	42
			1260	42		
			1278	42		
4.5	4.98		1520	50	50	50
			1512	50		
			1543	51		
4	4.48		1897	63	62	62
			1888	62		
			1918	63		
3.5	3.98		2364	78	77	77
			2306	76		
			2384	79		
3	3.48		3031	101	100	100
			3045	101		
			2997	99		
2.5	2.98		4056	135	135	135
			4028	134		
			4167	138		
2	2.48		5590	186	187	186
			5594	186		
			5683	189		
1.5	1.98		8098	269	267	267
			7935	264		
			8040	268		

Table 2: Count at different distance between source and detector

2.2 Analysing the data

As $N = N_0 e^{-\mu x}$, We get $ln(N) = ln(N_0) - \mu x$. Thus if we plot ln(N) vs thickness we sould get a straight line with slope $-\mu$. Data from table-1 is plotted. This graph is presented in figure-5.

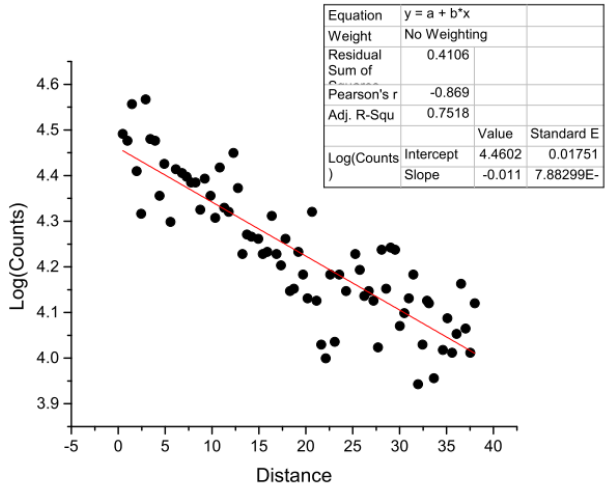


Figure 4: log of Counts per 30 seconds vs Thickness of the Aluminium Foil

As it can be seen from above figure slope is -0.011 thus the μ is 0.011 mm^{-1}. The intercept is 4.4602 thus $ln(N_0) = 4.4602 \Rightarrow N_0 = 86.66$ that is number of gamma rays emitted by the source in 30 second is 87. Half value thickness $t_{\frac{1}{2}} = \frac{ln2}{\mu} = 63.01mm$ that is after passing 63.01 in aluminium sheet number of gamma ray in the beam will get reduced to half of the original value.

To verify inverse square law we plot ln(N) vs distance. Data from tabl-2 is plotted. This graph is presented in figure-6.

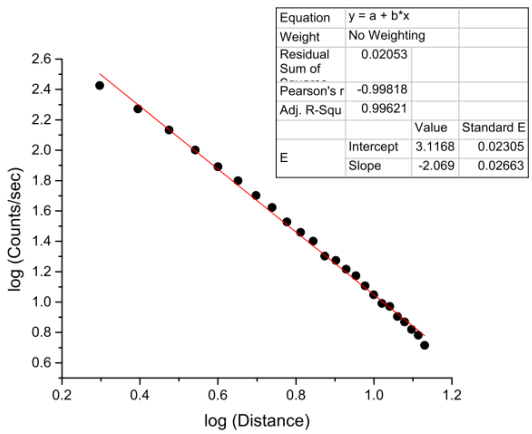


Figure 5: log of Counts per 30 seconds vs log of distance

As pointed out in section 1.2.5 we should get a straigt line with slope nearly -2. As shown in above figure the slope is -2.09 thus we verify the inverse square law.

3 conclusion

- We report the attenuation constant(μ) for aluminium to be 0.011$\pm$0.0078.
- We verify the inverse square law of radiation.